MIL-A-82654(OS)
31 January 1977
Superseding
(see section 6)

MILITARY SPECIFICATION

ALUMINUM POWDER, ATOMIZED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of atomized aluminum powder referred to herein as "the material".

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-1218	ACS Chemicals

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Maryland 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of the invitation for bids or the request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 34-72 — Chemical Analysis of Aluminum and Aluminum Based Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

3.1 Material. The material shall be a dull gray, atomized aluminum powder.

3.2 Chemical and physical properties. The chemical and physical properties shall be in accordance with TABLE I.

3.3 Workmanship. The material shall be uniform, free from contamination, foreign material, or any other defect that would prevent its use for the purpose intended.

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TABLE I. Chemical and physical properties

Property	Values	
	Minimum	Maximum
Free metallic aluminum, wt%	98.0	-
Volatiles, wt%	-	0.10
Iron as Fe, wt%	-	0.40
Oil and grease, wt%	-	0.10
Average particle diameter, µm	6.0	8.0
Weight mean diameter, µm	9.0	12.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)

b. Altitude: Normal ground

c. Vibration: None

d. Humidity: Room ambient to 95 percent relative, maximum

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all material manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time. Several batches, manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

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4.3.2 Sampling. Sampling for quality conformance inspection shall be in accordance with inspection level I of MIL-STD-105. The sample unit shall be one unit package or container of material. Each sample shall consist of sufficient material to perform the quality conformance tests as specified in 4.4.

4.4 Quality conformance inspection. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests and examination of 4.5. The acceptable quality level (AQL) shall be 2.5% defective. When specified in the contract (see 6.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests specified for each lot of material.

4.5 Tests. Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Visual examination. All samples shall be visually examined to verify the workmanship requirements.

4.5.2 Free metallic aluminum. Free metallic aluminum shall be determined in accordance with the following:

a. Reagents:

1. Sodium hydroxide solution, 5 percent (%).

2. Hydrogen gas.

b. Apparatus: Gas volumetric apparatus (eudiometer) as shown in Figure 1 including:

1. 1-liter (ℓ) leveling bottle or bulb.

2. Gas measuring burette 500 ml capacity. Graduated from 0 to 10 ml and 450 to 500 ml in 0.2 ml divisions.

3. 24/40 Standard Taper ground joint with stopcock.

4. 500 ml round bottom flask (reaction flask) with thermometer well. 35/20 ball and socket joint.

5. Small, glass covered magnetic bar enclosed in off-set of dropping device.

6. Thermometer, 0 to 100°C

7. Rubber tubing.

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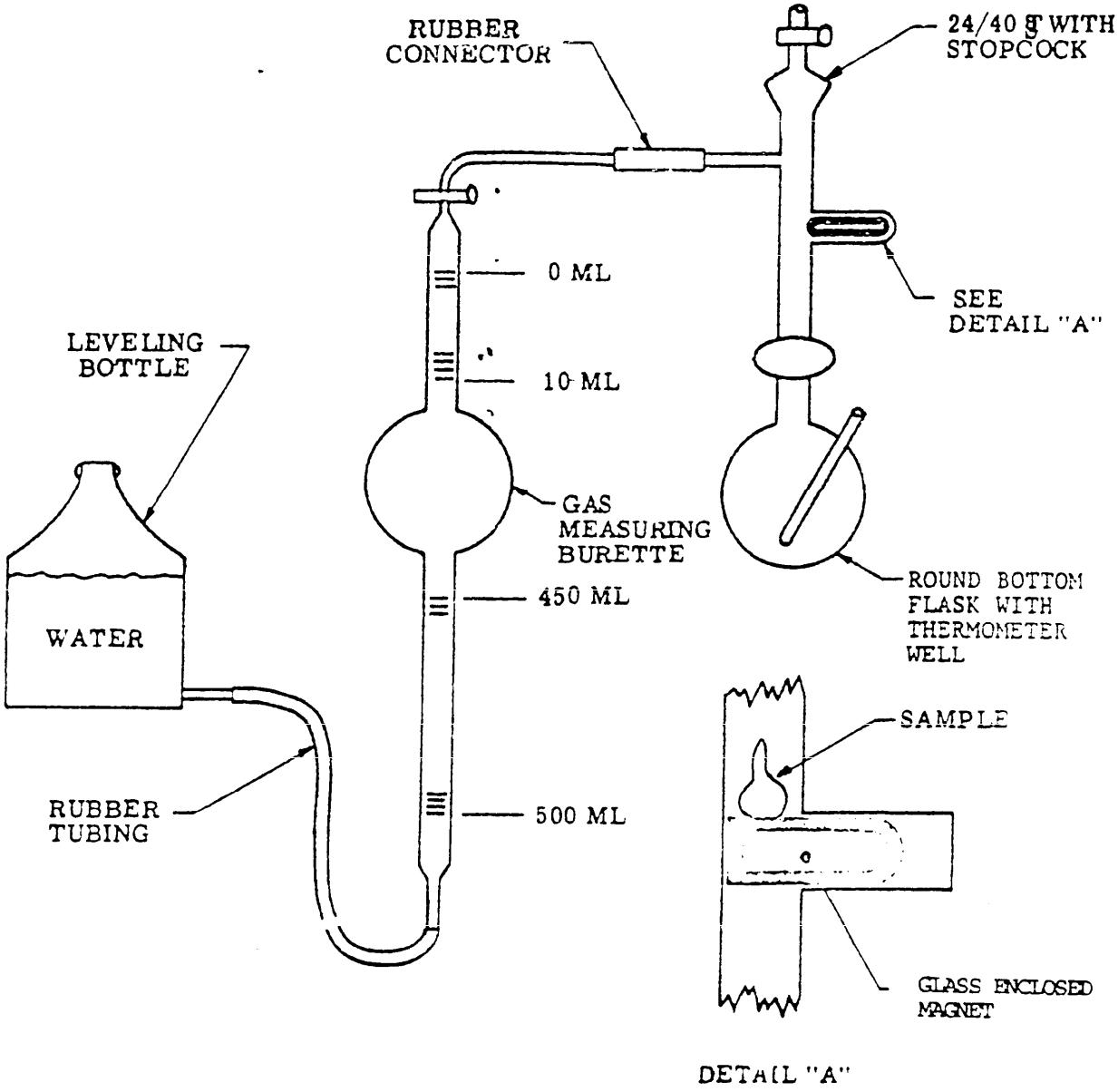


Figure 1. Schematic of gas volumetric apparatus for free metallic aluminum determination.

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c. Procedure: Weigh a 0.33 to 0.35 gram (g) sample to the nearest 0.1 mg, onto a tared 7 centimeter (cm) circle of filter paper. Fold the paper to contain the sample and prevent spillage until the sample is dropped into the sodium hydroxide solution. Remove the reaction flask from the apparatus and place the sample on the magnet in the drop device. Transfer 200 ml of the 5 percent sodium hydroxide solution to the reaction flask and saturate it with hydrogen gas. Re-attach the reaction flask to the apparatus, using a ball and socket pinch clamp and taking care to prevent the sample from falling into the alkali. Saturate the water in the leveling bottle with hydrogen gas. Bring the solutions to room temperature and record temperature. Open the stopcock at the top of the dropping device, raise the leveling bottle slowly until the meniscus in the burette and leveling bottle are exactly at the zero point of the burette. Close the stopcock and check the system for leaks by returning the leveling bottle to the table top for approximately 5 minutes. Raise the leveling bottle until the meniscus is level with the meniscus in the burette. Both will return to zero if no leaks are present. Return the leveling bottle to the top of the table and cause the sample to fall into the alkali. (This may be done by the use of a second magnet.) (See Figure 1, Detail A). After the initial reaction has subsided, shake the reaction flask several times, being careful not to cause a gas leak at the ground glass joints. Allow at least 20 minutes for completion of the reaction. Then bring the reaction flask to the starting temperature by immersing it in a water bath and maintain at this temperature for a minimum of 5 minutes. Adjust the leveling bottle until the meniscus of the leveling bottle and the burette are level and record the gas volume and barometric pressure

d. Calculate the free metallic aluminum as follows:

$$\text{Aluminum, wt\%} = \left[\frac{0.0002883V\ (P - p)}{W(273 + T)}\right] \times 100$$

where V = measured volume of gas, ml
P = barometric pressure, mm of mercury (Hg)
p = vapor pressure of water, mm Hg at T
T = temperature of system at equilibrium, ° C
W = weight of sample, g
0.0002883 = gas constant factor
273 = conversion from ° C to ° Kelvin (°K)

e. Report the results of a minimum of 2 determinations and their average.

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dures may be used by the contractor when they provide, as a minimum, the level of quality

4.5.3 Volatiles. Volatiles shall be determined in accordance with the following:

a. Procedure: Weigh approximately 5 g of sample, to the nearest 0.1 mg, into a tared moisture dish. Heat the dish and contents for a minimum of 4 hours at 105° ± 3° C, cool in a desiccator and reweigh.

b. Calculation: Volatiles wt%, $= \left[\frac{(A - B)}{A - C}\right] \times 100$

where: A = weight of moisture dish and sample, g
B = weight of moisture dish and dried sample, g
C = weight of dry moisture dish, g

c. Report the results of a minimum of 2 determinations and their average.

4.5.4 Iron content. Iron content shall be determined in accordance with 4.5.4.1 or 4.5.4.2. In the event of controversy, the results obtained by the method of 4.5.4.1 shall be determinative.

4.5.4.1 Preferred method for determing iron content. The preferred method for determination of iron content shall be in accordance with the Iron by the 1,10-Phenanthroline (Photometric) Method of ASTM E 34-72.

4.5.4.2 Alternate method for determining iron content. The alternate method for determination of iron content shall be in accordance with the following:

a. Reagents:

1. Concentrated hydrochloric acid.

2. Concentrated nitric acid.

3. Sulfuric acid solution, 50%.

4. Hydroxylamine hydrochloride solution, 10%, freshly made.

5. Sodium citrate solution, 25%.

6. O-phenanthroline solution, 0.25%.

7. Bright iron wire, primary standard grade.

8. Litmus paper.

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b. Apparatus:

1. Volumetric flasks - 100 ml, 500 ml, and one liter.

2. Pipets.

3. Beckman spectrophotometer, model DU, with matched one cm cells, set up for use with the tungsten lamp and red sensitive phototube, without filter.

c. Calibration (determination of absorptivity): Weigh 0.05 g of bright iron wire, to the nearest 0.1 mg, and place it in a one liter volumetric flask. Add 2 ml of concentrated hydrochloric acid and a few ml of hot distilled water. When all the iron is dissolved, fill the flask to volume and mix thoroughly. Transfer, by means of a pipet, 1, 3, 5, 7, and 9 ml aliquots of the standard iron solution to five 100 ml volumetric flasks and prepare a blank. Dilute each to about 70 ml with distilled water. Add 1 ml of hydroxylamine hydrochloride solution, mix well, and let the mixture stand for a minimum of 15 minutes. Add 10 ml of 0.25% o-phenanthroline solution, and mix well. Add 1 ml of 25% sodium citrate solution, dilute to volume, mix well and allow to stand for at least 30 minutes at not less than 20° C. Using the blank for comparison, determine the absorbence (a) of the sample solution at a wavelength of 510 nm and slit width of approximately 0.18 mm.

d. Calculation: Absorptivity, ml/g·cm = $\frac{A}{(L)\ (C)}$

where: C = concentration of iron in diluted aliquot, ml

A = absorbence (formerly referred to as absorbency)

L = length of light path through the cell, cm

e. Procedure: Weigh a 1 g sample, to the nearest 0.1 mg, into a 50 ml beaker. Transfer the sample quantitatively to a 500 ml volumetric flask. Wash the sides of the beaker and funnel with distilled water. Add 10.5 ml of concentrated hydrochloric acid and 5 ml of concentrated nitric acid and allow the sample to dissolve. (The reaction may be hastened by submerging the flask in a heated water bath but never heat on a hot plate). If the reaction becomes violent, spray with cold distilled water from a wash bottle. When the reaction is complete, dilute the solution almost to volume and allow the solution to come to room temperature. (This may be

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hastened by cooling the flask under tap water). When the solution is cool, dilute to volume with distilled water and invert the flask 20 times. To redilute the solution to a concentration such that the absorbence is between 0.05 and 10, pipet 50 ml of solution into a a 100 ml volumetric flask and dilute to 70 ml with distilled water. Prepare a blank using all reagents except the sample solution. Add 1 ml of hydroxylamine hydrochloride solution, mix well and let the mixture stand for a minimum of 15 minutes. Add 10 ml of 0.25% o-phenanthroline solution and mix well. Add 1 ml of 25% sodium citrate solution, dilute to volume, mix well and allow to stand a minimum of 30 minutes at not less than 20° C. Using the blank for comparison, determine the absorbence (A) of the sample solution at 510 nm and a slit width of approximately 0.18 mm.

f. Calculation: Iron as Fe, wt% = $\left[\frac{700\ A}{L\ K\ W}\right] \times 100$

where: A = absorbence of sample solution

K = absorptivity (from d above) ml/g·cm

W = sample weight, g

700 = concentration factor, ml

L = length of light path through the cell, cm

g. Report the results of a minimum of 2 determinations and their average.

4.5.5 Oil and grease. Oil and grease shall be determined in accordance with the following:

a. Reagent: Diethyl ether, absolute

b. Apparatus:

1. Soxhlet extractor.

2. 125 ml extraction flask, dried and weighed to ± 0.2 mg

c. Procedure: Weigh a 5 g sample, to the nearest 0.1 mg, and transfer into the thimble of the soxhlet extractor. Add approximately 100 ml diethyl ether to carry out the extraction and extract the sample on a steam bath for a minimum of 3 hours. Adjust the rate of heating so that the ether drops from the end of the condenser at the rate of 2 to 3 drops per second. Evaporate the ether from the extraction flask over a steam bath. Finally, dry the residue completely by heating to a constant weight (± 0.5 mg) in an oven at 105° ± 3° C.

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d. Calculation: Oil and grease, wt% = $\left[\frac{(A-C)}{B}\right]$ X 100

where: A = weight of residue and extraction flask, g

B = weight of sample, g

C = weight of extraction flask, g

e. Report the results of a minimum of 2 determinations and their average.

4.5.6 Average particle diameter. Average particle diameter shall be determined in accordance with the following:

a. Reagent: National Bureau of Standards (NBS) cement sample number 114.

b. Apparatus:

1. Fischer Sub-Sieve Sizer with accessories and instrument manual. Calibrated against the NBS cement sample number 114.

2. Paper discs.

c. Procedure: Check and adjust the level of the water in the standpipe and the liquid meniscus in the manometer tube. Lay a paper disc over one end of the sample tube, and push one of the porous plugs about 1.3 cm into the sample tube, with the perforated surface of the plug against the surface of the paper disc, forcing the paper to crimp around the edges and precede the plug into the sample tube. The sample tube is placed in a vertical position in a rubber support with the paper side of the plug up. Weigh 2.70 g of sample, to the nearest 0.01 g and with the aid of a powder funnel, completely transfer the sample into the sample tube, tapping the side of the tube to level and settle the powder. Lay a second paper disc over the sample tube, and force the other porous plug and paper disc down into the sample tube. Place the sample tube on the brass post beneath the rack and pinion with the lower plug in contact with the upper end of the brass post. Pack the sample to a minimum porosity by turning the pinion knob manually to produce a maximum pressure on the sample. Shift the calculator chart laterally until the tip of the pointer just coincides with the sample height curve on the chart. The chart should not be moved after this setting until the

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determination is finished. (If the sample has a porosity greater than 0.8 or less than 0.4, adjust the sample weight to give an "apparent porosity" between these limits. Reweigh and repack a new sample, if the porosity is out of the 0.4 to 0.8 porosity limits.) Mount the sample tube between the rubber-cushioned supports and clamp the upper cap down so an air-tight seal is obtained at both ends. Adjust the pressure control knob until the bubbles rise in the standpipe at a rate of 2 to 3 bubbles/second. If the average particle diameter falls in the range of 0.2 to 20.0 µm, turn the range control indicator to the right. If the average particle diameter falls in the range of 20.0 to 50.0 µm, turn the range control indicator to the left. After the liquid level in the manometer tube has risen to a constant height, the rack is turned up until the upper edge of the cross bar coincides with the liquid meniscus in the manometer. The particle size is indicated by the location of the tip of the pointer in relation to the curves on the calculator chart. Read the chart directly, if the average particle diameter falls in the 0.2 to 20.0 µm range. If the average particle diameter falls between 20.0 and 50.0 µm, multiply the chart readings by two. When the porosity has been found to be greater than 0.8 or less than 0.4 and an adjusted weight has been used, use the following calculation to obtain a corrected average particle diameter.

d. Calculation: Corrected diameter, $\mu m = d_a \frac{m}{p} \left(\frac{P}{1 - \frac{m}{p}(1 - P)} \right)^{3/2}$

where: d_a = apparent diameter, µm (as read)
m = mass, g
p = density, g/cm^3
P = porosity (the ratio of the air space in the sample to the total volume occupied by the packed sample)

e. Report the results of a minimum of 2 determinations and their average.

4.5.7 Weight mean diameter. Weight mean diameter shall determined in accordance with the following:

a. Reagent: Nitrogen, prepurified.

b. Apparatus:

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1. Sharples Micromerograph, type XC.
2. Templates, 6 inches/minute and 6 inches/hour.
3. Stopwatch.

c. Cleaning procedure: Clean the micromerograph in accordance with the manufacturers instructions.

d. Instrument settings: Set the instrument for the aluminum in accordance with TABLE II.

TABLE II. Instrument settings.

Deagglomerator opening (µm)	Nitrogen pressure (psi)	Sensitivity position	Approximate sample size (mg)
250	30	1	40

e. Procedure:

1. Remove the sample holder by releasing the clamping handle and pulling straight out.
2. With a micro-spatula, scoop a small portion of the material into the sample holder. (See TABLE II for approximate sample size.)
3. Replace sample holder into the feed chamber and clamp it into position.
4. Set deagglomerator opening, feed pressure and sensitivity (see TABLE II).
5. Normally the power to the micromerograph is left on. This keeps the machine in a ready condition and avoids warm-up time.
6. With the recorder switch on, push down on the chart speed lever. The down position drives the chart at 6 inches/minute and the up position drives the chart at 6 inches/hour.
7. With the chart moving at 6 inches/minute move the recorder pen to the zero position on the chart, using the zero adjuster at the bottom of the recorder.

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8. Press the left hand charging button for 5 seconds (s) filling the pressure chamber.

9. Press the right hand firing button and hold down for 5 seconds. This discharges the powder into the sedimentation column.

10. After the powder has been discharged into the column, let 225 s elapse and then change the chart speed to 6 inches/hour by pushing the chart speed level upward.

11. Allow the run to continue until the pen deflection becomes constant. The micromerograph run is then completed.

12. Turn off the recorder and open the sedimentation column by turning the seal operating handle counterclockwise. Place the balance pan protector disc over the balance pan.

13. If a total defelction of 60 is not obtained, the run is discarded and the sample is rerun.

14. Remove the chart from the recorder and identify the chart as to material, lot number, instrument settings, date and any other pertinent information.

f. Data reduction and calculations:

1. Lay the template marked 6 inches/minute over the chart, aligning it with the change of speed point and align the lines parallel to the sides of the template with the 0 to 100 scale on the outer edge of the recorder chart for the full length of the template. Read the chart deflection at the intersection of the template arc lines and the chart curve, recording the data on the calculation sheet in column 3. See TABLE III.

2. Lay the template marked 6 inches/hour over the chart aligning it with the change of speed point and align the lines parallel to the sides of the template with the 0 to 100 scale on the outer edge of the recorder chart for the full length of the template. Read the chart deflection at the intersection of the template arc lines and the chart curve, recording the data on calculation sheet in column 3.

3. Normalize all figures in column 3 to 100 percent by dividing each by the last figure in column 3, and multiplying by 100 to convert to percentage, and record in column 4.

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4. Column 2 has been calculated by dividing each figure in column 1 by the square root of the density of aluminum.

5. Subtract each entry in column 4 from 100 and list the result in column 5.

6. Plot cumulative wt% less than diameter (column 5) as the ordinate (arithmetic scale) against diameter, in µm, (column 2) as the abcissa (log scale) on 3 cycle semi-log graph paper. Identify the plot as described in e. 14.

7. The weight mean diameter, in µm, is reported as the intercept of the 50 percent line and the curve plotted in 6. above.

4.6 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of section 5.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging, and packing shall be level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of aluminum powder shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation and shall be suitable for indoor storage at 32°C (90°F) maximum for a minimum of 2 years.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to the following information:

a. Title, number and date of this specification

b. Manufacturer's name and location

c. Material trade name

d. Net weight

e. Lot number, batch number(s), and date of manufacture

f. Storage conditions

g. Contract or purchase order number

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TABLE III. Calculation sheet, weight mean diameter of aluminum powder.

1	2	3	4	5
$d\sqrt{p}$	Diameter,	Chart deflection	Cumulative % greater than	Cumulative % less than
269	164			
180	110			
150	91.5			
120	73.2			
90	54.9			
75	45.7			
66	40.2			
60	36.6			
54	32.9			
48	29.3			
42	25.6			
36	22.0			
30	18.3			
27	16.5			
24	14.6			
21	12.8			
20	12.2			
19	11.6			
18	11.0			
17	10.4			
16	9.76			
15.1	9.21			
14	8.54			
12	7.32			
10	6.10			
8	4.88			
6	3.66			
5	3.05			
4	2.44			
3	1.83			
2.5	1.52			
2	1.22			

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6. NOTES

6.1 Intended use. Aluminum powder in accordance with this specification is intended for use as an ingredient in a solid propellant for the CCU-22/A impulse cartridge and other cartridge devices.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification

b. Quantity required

c. Place of delivery

d. Inspection conditions when other than as specified (see 4.2)

e. Lot size if other than as specified (see 4.3.1)

f. Packaging requirements if other than as specified (see 5.1)

g. Size of container required (see 5.1)

h. Any special markings required (see 5.2)

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID *
4.4	Quality conformance inspection data	-

*DID's (Data Item Description/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List), Such data will be delivered as identified on completed (numbered) DID's when specified on DD Form 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is Reynolds 1-131, manufactured by Reynolds Metals Company, P. O. Box 1057, Birmingham, Alabama 35201. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

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6.4 Supersession information. MIL-A-82654 is intended to be used in lieu of AS 3051 (Code Ident 30003) dated 15 November 1974 for Naval Sea Systems Command procurement.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Project Number:
6810-NA28

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977 703-020/1925

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-A-82654(OS) "Aluminum Powder, Atomized"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

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